

A DINAMIC TRIVIA GAME, BASED IN DESIGN PATTERNS

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Abstract

The goal of this project is to create a trivia game, where users can play and earn points for correct answers, this is why this group has decided to use the best options to solve this problem, these can be the separation by modules and the oob, followed the SOLID principles, and maintained some good practices, because the main objective is to make users have fun.

Introduction

The development of the trivia game was based on object-oriented programming (OOP), which allowed for the creation of a modular, clear and reusable system. SOLID principles were applied to ensure flexible, decoupled and easily extendable code, while the use of design patterns optimized the solution. Separating the code into modules facilitated collaborative work and the incorporation of new functionalities without compromising the stability of the system. This structured approach favored the efficiency, quality and sustainability of the project, allowing for simpler management and more agile development.

Goal

The team's goal is to deliver an application that meets all the user stories collected, delivering a project with code that, in addition to being easy to understand and explain for anyone, is scalable and maintainable over time.

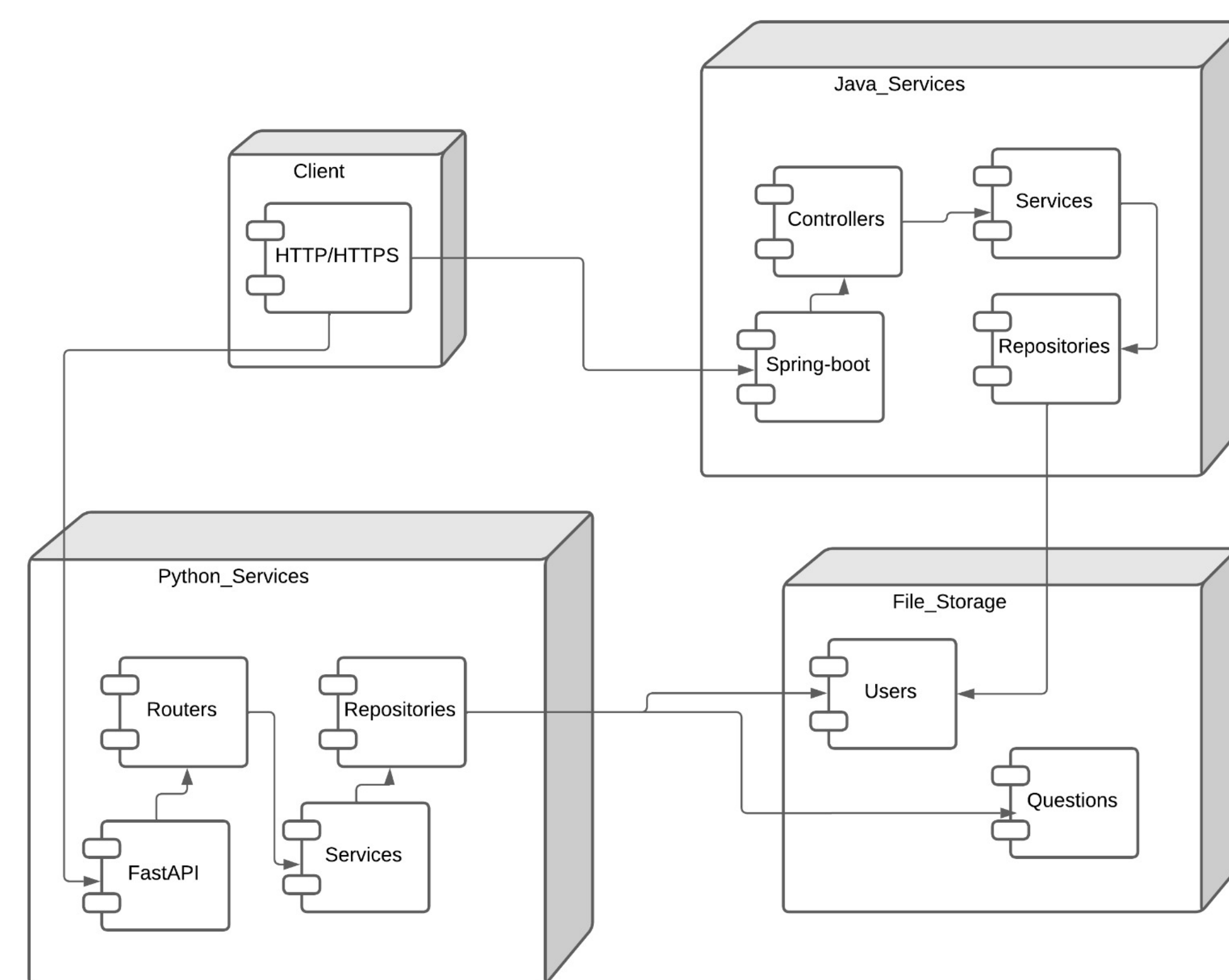


Figure 1: Deployment diagram.

Results

The team thoroughly reviewed and tested each function during development, performing unit tests on key modules to verify their correct behavior and detect errors early. Attention was given to ensuring that the system met the user stories defined at the start of the project. This approach resulted in a fully functional, tested solution that aligned with the initial objectives, making the trivia game stable and operational for users.

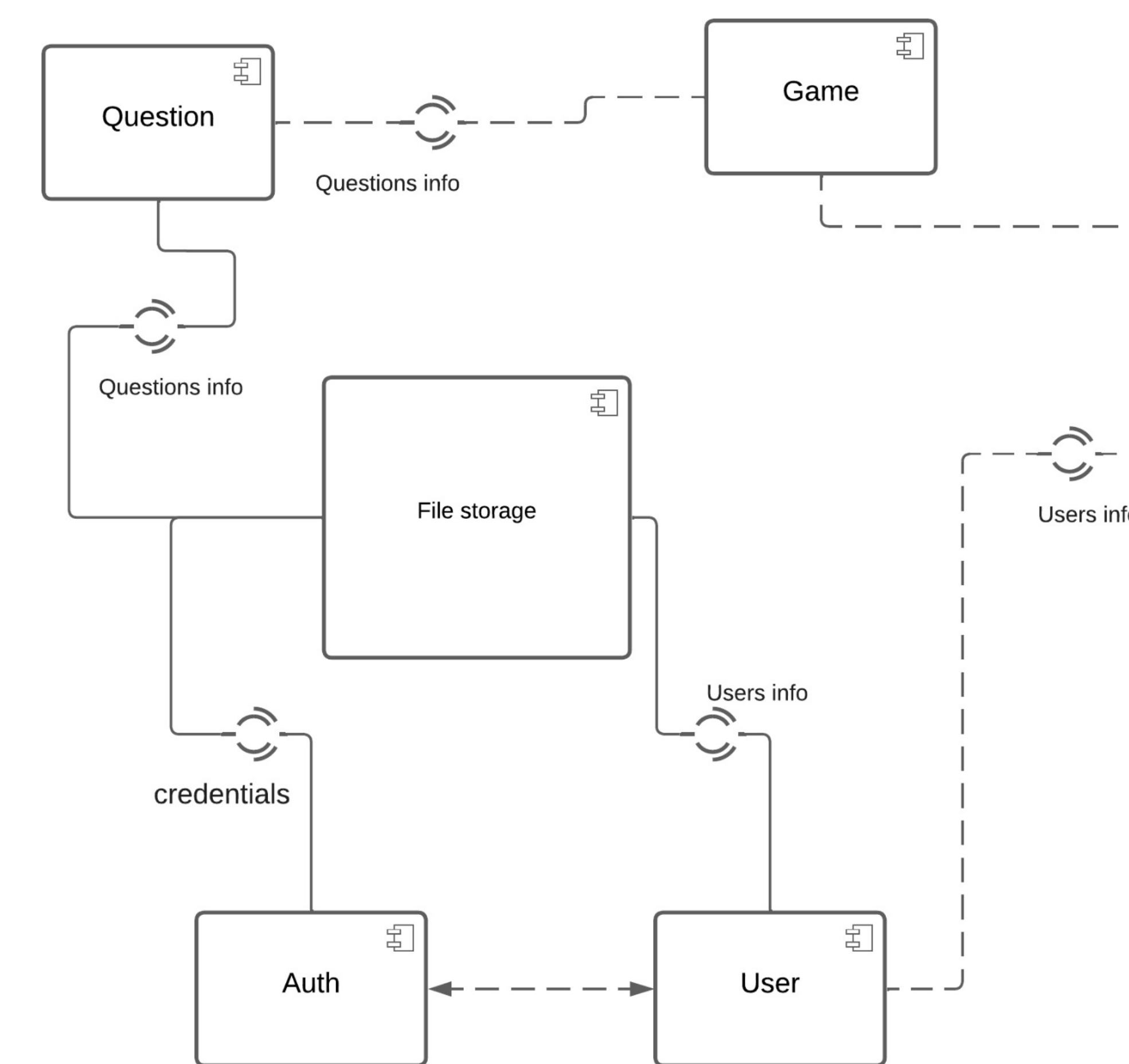


Figure 2: Component diagram.

Conclusion

The project highlighted the importance of a well-structured design for creating a functional and efficient system. Using object-oriented programming (OOP), a modular and scalable trivia game was developed, with clear responsibilities for each component. SOLID principles and design patterns improved code organization and future maintenance. Modularization allowed independent development and testing of each part, enhancing collaboration and speeding up the process. The project provided valuable technical insights and reinforced best development practices.

References

- [1] E. Gamma, R. Helm, R. Johnson, and J. Vlissides, *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley, 1994. Available on: <https://refactoring.guru/design-patterns>.
- [2] R. C. Martin, *Design Principles and Design Patterns*. 2000. Available on: https://web.archive.org/web/20110720091203/http://www.objectmentor.com/resources/articles/Principles_and_Patterns.pdf.
- [3] JSON.org, *Introducing JSON*. Available on: <https://www.json.org/json-en.html>.
- [4] S. Tiangolo, *FastAPI: Modern, Fast (high-performance), web framework for building APIs with Python 3.6+*. Available on: <https://fastapi.tiangolo.com/>.
- [5] Software Testing Fundamentals, *Unit Testing: Introduction to Unit Testing*. Available on: <https://softwaretestingfundamentals.com/unit-testing/>.